Signal-Worker Boundary Confinement: A Corrected Ontology of Surface vs Bulk Transport

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Abstract

The Signal-Worker (S-W) ontology — boson = *signal* (the delocalized field instruction), fermion = *worker* (the localized state that performs work) — is the QNFO corpus's proposed decomposition of the wave–particle duality fog. This paper delivers the red-team-hardened correction of that ontology's boundary-confinement reading, following a 3-slot adversarial audit (2026-08-14) that found 5 HARD defects. The correction is a taxonomy, not a renaming: only *mode-confinement* phenomena (topological insulators, the quantum Hall effect, and the non-Hermitian skin effect) confine fermionic transport to the material boundary; the Meissner effect and the AC skin effect expel the *field/current density* while the electrons (the "workers") continue to flow through the bulk. The mapping also fails on composite bosons (Cooper pairs — two fermions condensing into one boson) and self-conjugate particles (Majorana zero modes), and the corpus's own flagship substrate (TaAs, a Weyl semimetal with a gapless conducting bulk) contradicts the "worker excluded from the bulk" claim. Every claim carries an epistemic label and a falsifiability condition; the ontology itself is explicitly labeled an unconfirmed internal proposal.

v0.2 (2026-08-16). This newversion adds a classical-electrodynamics companion section (§3.6: bundling practice, skin-depth thresholds, and the quantum origin of the bulk–insulator distinction) and completes the deposited source set with `references.bib` and `citation-audit.md`. No changes to the v0.1 taxonomy or falsifiability register.

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1. Introduction and Positioning

The Signal-Worker ontology originates in five QNFO corpus records: *Unifying Photosynthetic Energy Transduction and Ambient Superconductivity* [7], *Structural versus Driven Quantum Coherence* [8], *Quantum Architectonics* [9], *Gauge-Invariant Field Theory of Signal-Worker Interactions* [10], and *Quantum Abacus* [11]. It proposes that bosons (photons/phonons) act *strictly as informational signals* directing localized fermions (electrons/excitons) to perform work, and that this functional decomposition is "non-dualistic."

A 3-slot CMD RED TEAM SUB (Accuracy / Completeness / Dependency, 2026-08-14) audited the ontology's central reading — *"the signal orders where the worker may act; in topologically-protected phases the worker is excluded from the bulk and confined to the boundary."* Verdict: the underlying physics claims are accurate (Accuracy: 0 HARD), but the unifying reading is overgeneralized (Completeness: 5 HARD) and the corpus carries a terminology collision (Dependency: LCI used for two different concepts). This paper formalizes the corrected position.

Scope. This paper does NOT re-derive the S-W ontology or the topological-material corpus (see the deep-due-diligence report, companion artifact). It corrects the boundary-confinement claim, the composite-particle mapping, and the epistemic status of the ontology, and it fixes the LCI terminology collision. v0.2 adds §3.6 — the classical-EM companion that applies this correction to engineering practice — plus the two missing provenance files.

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2. The Signal-Worker Ontology, Stated

Role	Carrier	What it does	Corpus label
Signal	boson (photon / phonon)	carries the *instruction* — a delocalized field modifier	information
Worker	fermion (electron / exciton)	performs the *work* — a localized state vector	action

Statement 1 [MAP — interpretive, proposed]. Wave–particle duality is functionally decomposable into a bosonic *instruction* role and a fermionic *action* role in driven non-equilibrium systems. *Status: internal proposal; unconfirmed; ambient superconductivity has never been achieved [SOFT-9].* The framing is *hierarchical* (the signal orders the worker), so it is not non-dualistic in the strict sense — it privileges one pole.

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3. The Established Physics: Surface vs Bulk — a Taxonomy

The counterintuitive surface-vs-bulk phenomena the ontology cites are real, but they are **mechanistically heterogeneous**. They split into three classes:

3.1 Mode confinement (topological insulators) — TERRITORY

A 3D Z_2 topological insulator has a gapped (insulating) bulk and hosts gapless, metallic *surface* states [1,2]. The surface states are protected by time-reversal symmetry and the bulk Z_2 index via the bulk–boundary correspondence. **Electron transport is genuinely confined to the boundary.** This is a *mode-confinement* phenomenon: the bulk does not conduct, the surface does, and the surface modes are spin-momentum locked (helical).

3.2 Mode confinement (quantum Hall effect) — TERRITORY

In a 2D electron gas at strong magnetic field, the quantized Hall conductance is carried by **chiral edge channels**; the bulk is inert [2]. Again, transport is genuinely boundary-localized (chiral, in this case).

3.3 Field expulsion (Meissner effect) — TERRITORY

A superconductor actively **expels** the magnetic field: B ≈ 0 in the bulk, with the field decaying over the London penetration depth (λ_L ≈ 10–100 nm) [2]. This is *active expulsion*, distinct from a perfect conductor's flux-trapping. **Crucially, the electrons do NOT leave the bulk** — they flow through it as a dissipationless supercurrent. The London depth is the *field's* decay length, not an electron-localization length. The "worker" stays; the "field" leaves.

3.4 Current-density redistribution (AC skin effect) — TERRITORY

At high frequency, AC current density crowds toward the conductor surface over the skin depth δ ∝ 1/√f. This is a *current-density redistribution* within the conductor — **no particle confinement**, no boundary-localized eigenmodes.

3.5 Mode collapse (non-Hermitian skin effect) — TERRITORY (external, recent)

Under non-Hermiticity (gain/loss or asymmetric hopping), the **non-Hermitian skin effect (NHSE)** drives *all* bulk eigenmodes to collapse onto the boundary (Yao–Wang 2018) [3]. This is the literal "bulk → boundary" analog and the most on-point mode-confinement phenomenon for the ontology's vocabulary — and it is absent from the S-W corpus [H4].

3.6 Engineering companion: bundling, skin-depth thresholds, and the quantum bulk–insulator boundary [v0.2]

A recurring misreading holds that conductors are *bundled* because of the skin effect. The engineering record says otherwise — and the distinction is a live instance of the category discipline of §4. [TERRITORY]

Bundling practice. Three distinct practices are conflated under "bundling":

- **Stranded conductors** (ordinary copper cable) exist for mechanical flexibility. At 50/60 Hz the skin depth in copper is $\delta = \sqrt{2\rho/\omega\mu} \approx 8.5$ mm, so stranded wires below ~17 mm overall diameter conduct through essentially the full cross-section at mains frequency.

- **Bundled conductors on high-voltage lines** (2–4 subconductors per phase, usually aluminum conductor steel-reinforced — ACSR, not copper) exist primarily to suppress **corona discharge** by reducing the surface electric-field gradient. Electrically, bundling lowers the series inductance (larger effective geometric mean radius) and raises the shunt capacitance (larger equivalent radius), which raises the surge-impedance loading and the power-transfer capability.
- **Litz wire** (individually insulated strands) is the only bundling scheme whose purpose *is* the skin/proximity effect — and it exists for high-frequency use, not mains.

So the AC skin effect is real, but almost none of the wire we see is bundled because of it. The error is a category slip: a *field/current-density* phenomenon (skin effect) is misattributed as the motive for a *structural* practice (bundling) — the same slip the corrected taxonomy of §4 exists to prevent.

Where quantum effects actually enter the bulk–insulator boundary. The conductor/insulator distinction itself is quantum mechanical: copper conducts because its band structure provides a partially filled band of delocalized Bloch states, while an insulator presents a filled valence band plus a gap. The AC skin effect, by contrast, is classical Maxwell electrodynamics — no quantum input is required at macroscopic scales. Quantum physics becomes explicit for surface-vs-bulk questions when a length scale approaches the Fermi wavelength ($\lambda_F \approx 0.46$ nm in copper) or the electron mean free path (~40 nm at room temperature): conductance quantization in multiples of e^2/h, ballistic transport, quantum confinement, and — genuinely quantum and genuinely boundary-localized — the topological surface states of §3.1, protected by bulk topology. The skin effect and the topological boundary are therefore *not* the same kind of "surface": the former is a classical field redistribution inside a conductor; the latter is a mode-confinement effect. The §4 category distinction survives the nanoscale.

S-W reading. In Signal-Worker vocabulary: under the AC skin effect, the *signal* (the electromagnetic field) is redistributed toward the surface while the *workers* (the conduction electrons) remain distributed through the bulk; in a topological insulator, the boundary *modes* carry the work. Nothing in this companion section changes the v0.1 taxonomy — it supplies the engineering and quantum-threshold context in which that taxonomy is applied.

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4. The Category Error and Its Correction [H1]

Finding [HARD]. The S-W reading "the worker is excluded from the bulk and confined to the boundary" conflates **field/current-density expulsion** (Meissner, skin effect) with **mode confinement** (topological insulator, quantum Hall). They are different physical mechanisms with different consequences for where the "worker" may act.

Correction (this paper's primary claim):

Class	Phenomena	What is at the boundary	Does the "worker" (electron) leave the bulk?
Mode confinement	TI surface states, QH edge	Electron transport modes	**Yes** (bulk gapped/inert)

Class	Phenomena	What is at the boundary	Does the "worker" (electron) leave the bulk?
	channels, NHSE		
Field expulsion	Meissner effect	The B field	**No** (electrons flow through bulk)
Current-density redistribution	AC skin effect	The current density	**No** (redistribution, not confinement)

Statement 2 [MAP — to be defended]. The boundary-confinement reading of the S-W ontology is valid **only** for the mode-confinement class. A corrected framing must distinguish field-expulsion from mode-confinement. *Falsifiability condition C2: the corrected taxonomy is disconfirmed if it yields no new observable consequence beyond the standard bulk–boundary correspondence — i.e., if it is pure relabeling (the criterion the Prime Valuation Depth follow-on applied to itself).*

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5. The Composite-Boson Problem [H3]

Finding [HARD]. The ontology's flagship example is superconductivity, yet the superconducting charge carrier is a **bosonic Cooper pair** — two fermionic "workers" condensing into a "signal-like" boson. The mapping `boson = signal / fermion = worker` is **not closed under composite particles**:

- Cooper pairs (two fermions → one boson): the superconducting "worker" is a boson.
- Phonons: bosons that serve as both the pairing *glue* (the signal) and a lattice excitation — the instruction/action split blurs.
- Majorana zero modes (self-conjugate, neither boson nor fermion): fall outside both labels. The corpus itself works this thread (ZBW-Majorana, [14]).

Correction. The functional split is a *regime-specific interpretation*, not a universal classification. It must explicitly exclude composites and self-conjugate particles.

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6. Spin-Statistics Engagement [SOFT-6]

Finding. "boson = signal / fermion = worker" is a teleological gloss over the real boson/fermion distinction — the **spin-statistics theorem** (integer vs half-integer spin). The ontology re-describes a kinematic fact without deriving a new observable.

Correction. The corrected framing must engage spin-statistics directly: the *kinematic* distinction (integer/half-integer spin → symmetric/antisymmetric statistics) is the ground truth; the *functional* signal/worker roles are a valid interpretation only where field modifiers and localized carriers are mechanistically distinct (driven non-equilibrium regimes).

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7. The Weyl-Semimetal Counterexample [H2]

Finding [HARD]. The corpus's own flagship substrate is **TaAs, a Weyl semimetal** — with a **gapless, conducting bulk** and Fermi-arc *surface* states (Quantum Abacus [11]; Xu et al. 2015 [5]; Wan et al. 2011 [4]). The corpus additionally contains *Hamiltonian Engineering of Topological Deconfinement in Weyl Semimetals* [12], which already engages Weyl physics directly. Weyl semimetals are **not** Z_2 topological insulators: the bulk conducts. Therefore "the worker is excluded from the bulk" is **false for the corpus's own material**.

Correction. The paper must distinguish: (a) gapped-bulk mode confinement (TI/QH), (b) gapless-bulk Weyl/Dirac semimetals (Fermi arcs — boundary states *on top of* a conducting bulk). The S-W vocabulary may describe Fermi arcs as boundary-*enhanced* work, but not as bulk-*excluded* work.

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8. The LCI Acronym Collision [H5]

Finding [HARD — dependency]. The corpus uses **LCI** for two different concepts:

Record	LCI =
Gauge-Invariant Field Theory [10] (10.5281/zenodo.18466522)	**Logical Cloning Prohibition** (Ward identity of the Signal gauge field)
Structural vs Driven [8] (10.5281/zenodo.18441402); Quantum Architectonics [9] (10.5281/zenodo.18515458)	**Lossless Complexity Index**

Correction. This paper uses **LCI** only for *Logical Cloning Prohibition* [10] and spells out *Lossless Complexity Index* in full wherever it appears [8,9]. Any cross-corpus citation must disambiguate.

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9. The Corrected Signal-Worker Framing

The corrected framing is a **three-part statement**:

1. **[TERRITORY]** The boson/fermion distinction is the spin-statistics theorem (kinematic). Surface-vs-bulk transport splits into mode confinement (TI, QH, NHSE — boundary-localized transport, gapped/inert bulk), field expulsion (Meissner — field leaves, electrons stay), and current-density redistribution (skin — no confinement).
2. **[MAP]** The S-W "instruction vs action" split is a valid *functional* interpretation in driven non-equilibrium regimes, and a valid *mode-confinement* statement only for the mode-confinement class. It is not a universal decomposition: it fails on composite bosons (Cooper pairs), self-conjugate particles (Majorana zero modes), and gapless-bulk Weyl semimetals.
3. **[EPISTEMIC]** The ontology is an **unconfirmed internal proposal**. Ambient superconductivity has never been achieved; the Ward-identity no-cloning derivation (LCI [10]) is flagged speculative and must be independently reproduced. Boundary

confinement is a *transport* statement — it does NOT imply topological-QC fault tolerance at nonzero temperature (the corpus's own evidence synthesis refutes the FCI alternative via thermal anyon proliferation; see deep-due-diligence companion).

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10. Falsifiability Register

#	Claim	Type	Falsifiability condition	Status
C1	TI/QH confine fermionic transport to the boundary; Meissner/skin expel the field/current	established	—	established
C2	The corrected taxonomy (field-expulsion vs mode-confinement) carries content beyond relabeling	MAP	no new observable consequence → relabeling	OPEN
C3	LCI (Logical Cloning Prohibition) as a Ward identity, exponential-in-N scaling	MAP-speculative	independent derivation + reproduction fails	OPEN, flagged
C4	Composite bosons (Cooper pairs) break the boson=signal/fermion=worker mapping	MAP	a closed mapping under composites exists	OPEN

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11. Conclusion

The counterintuitive surface physics of topological conductors is real and correctly described — but it is **narrower** than the Signal-Worker framing claimed. "Electron confined to the surface" holds for topological insulators, the quantum Hall effect, and the non-Hermitian skin effect (gapped/inert bulk → boundary transport). It does *not* hold for the Meissner effect or the AC skin effect (field/current expulsion — electrons still flow through the bulk), and it does *not* hold for the corpus's own Weyl-semimetal substrate (conducting bulk with Fermi arcs). The Signal-Worker ontology, corrected, is a *regime-specific functional interpretation*, explicitly unconfirmed, with its composite-particle, spin-statistics, and terminology gaps closed. The physics survives the audit; the ontology, corrected, survives it too — as a falsifiable MAP, not as established doctrine.

Declarations

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availability. No code was required for this work. **Ethics approval.** Not applicable. **Preprint policy.** This manuscript is posted as a preprint; it has not been submitted to any journal.

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